Formato dos dados

1. Os diferentes formatos de sequência
2. Escore de qualidade e FASTQ
3. Exemplo de um arquivo Illumina



Formato dos dados : os diferentes formatos de sequências

Mais populares:

- FASTA (.fasta, .fa, .fna, .fsa)
- FASTQ (.fastq, .fq)
- SFF (SequenceFlowgramFormat: .sff)
- CSFASTA (Color-Space FASTA: .csfasta)
- SRA (SequenceReadArchive: .sra)
- SRF (SequenceReadFormat: .srf)
- ABI (ABI sequencerfile: .ab, .abi ou .ab1)
- SCF (SequenceChromatogramFormat: .scf)
- PHD (PHreD: .phdou .phd.1)
- …

O formato Microsoft WORD (.doc) não é um formato de sequência !!

Formato dos dados : os diferentes formatos de sequências

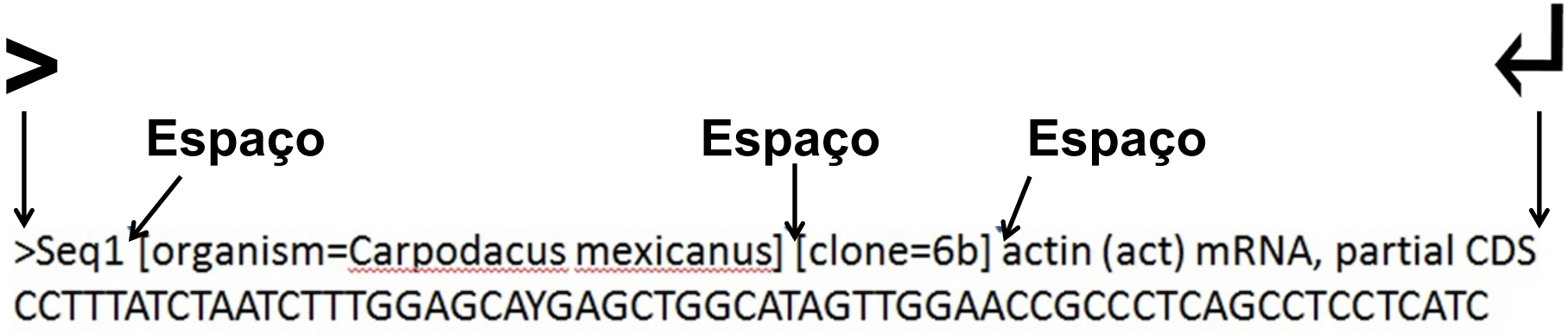
Mais populares:

- FASTA(.fasta, .fa, .fna, .fsa)
- FASTQ(.fastq, .fq)
- SFF(SequenceFlowgramFormat: .sff)
- CSFASTA(Color-Space FASTA: .csfasta)
- SRA(SequenceReadArchive: .sra)
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- …

O formato Microsoft WORD (.doc) não é um formato de sequência !!

Formato dos dados : o formato FASTA

FASTA



NomenclaturaIUB/IUPAC

A adenosine	C cytidine	G guanine
T thymidine	N A/G/C/T(any)	U uridine
K G/T(keto)	S G/C(strong)	Y T/C(pyrimidine)
M A/C(amino)	W A/T(weak)	R G/A(purine)
B G/T/C	D G/A/T	H A/C/T
V G/C/A	–gap of indeterminate length	

Formato dos dados : o formato FASTA

Pode conter sequências de aminoácidos também:

>MCHU–Calmodulin–Human, rabbit, bovine, rat, and chicken
ADQLTEEQIAEFKEAFSLFDKDGDGTITTKELGTVMRSLGQNPTEAELQDMINEVDADGNGTID
FPEFLTMMARKMKDTDSEEEIREAFRVFDKDGNGYISAAELRHVMTNLGEKLTDEEVDEMIREA
DIDGDGQVNYEEFVQMMTAK*

>gi|5524211|gb|AAD44166.1| cytochrome b [Elephasmaximus maximus]
LCLYTHIGRNIYYGSYLYSETWNTGIMLLLITMATAFMGYVLPWGQMSFWGATVITNLFSAIPYIGTNLV
EWIWGGFSVDKATLNRFFAFHFILPFTMVALAGVHLTFLHETGSNNPLGLTSDSDKIPFHPYYTIKDFLG
LLILILLLLLLALLSPDMLGDPDNHMPADPLNTPLHIKPEWYFLFAYAILRSVPNKLGGVLALFLSIVIL
GLMPFLHTSKHRSMMLRPLSQALFWTLTMDLLTLTWIGSQPVEYPYTIIGQMASILYFSIILAFLPIAGXIENY

Formato dos dados : Escore de qualidade e FASTQ

- O phred lê dados de sinais de fluorescência oriundos de sequenciadores automáticos de DNA
- Define as bases (*base-calling*)

Escore de qualidade Phred: $Q = -10 \log_{10} P$

Probabilidade de erro

Escore de qualidade Phred	Probabilidade de base-calling errado	Precisão do base-calling
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.99%
50	1 in 100000	99.999%

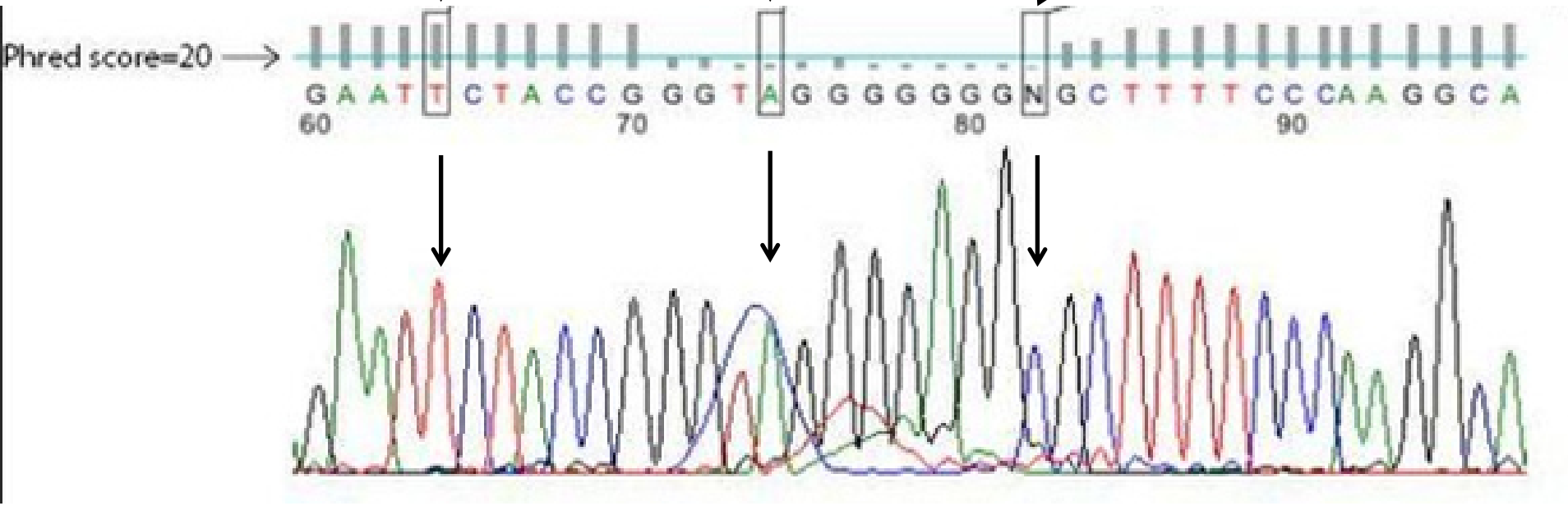
Formato dos dados : Escore de qualidade e FASTQ

Exemplo com sequenciamento Sanger

Phred = 50
A probabilidade que essa base é uma timina = 99.999 %

Phred = 10
A probabilidade que essa base é uma adenina = 90 %

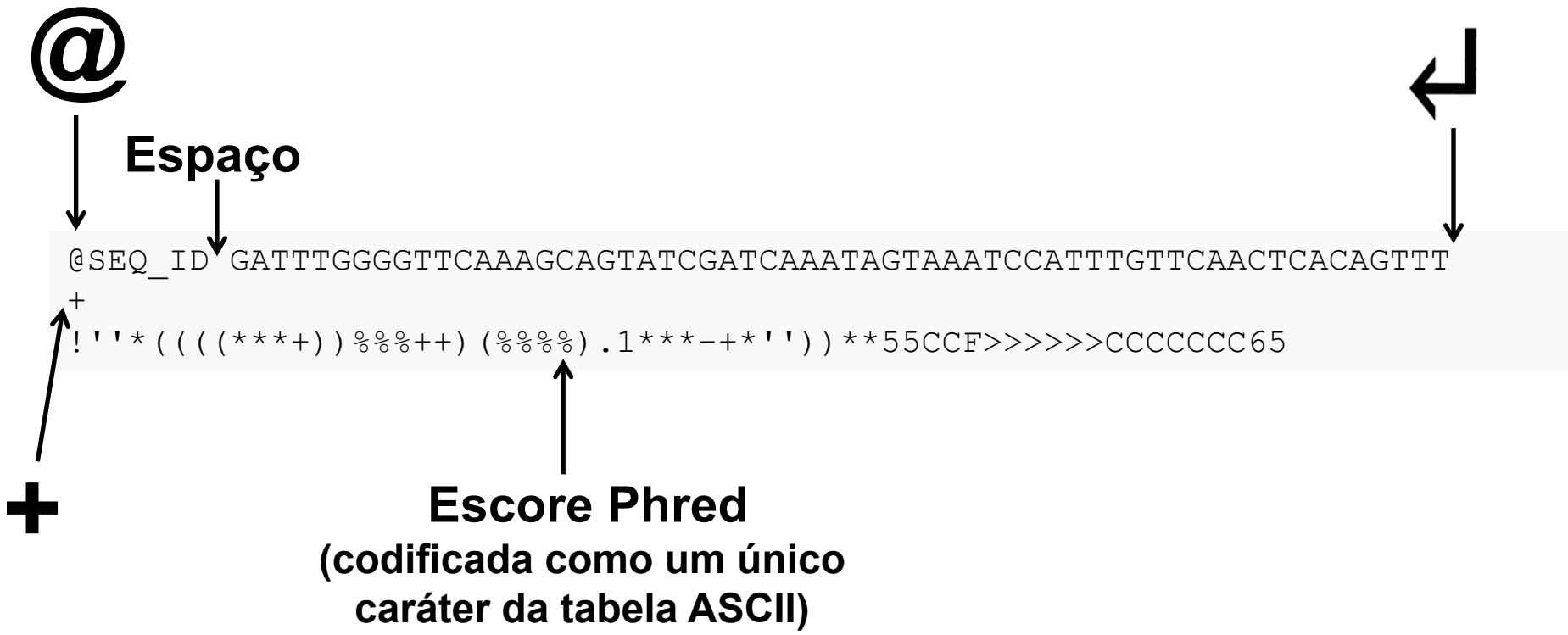
Phred não calculado
O sequenciador não determinou qual é a base (N)



Formato dos dados : Escore de qualidade e FASTQ

Formato FASTQ

Fasta + escore Phred



Formato dos dados : Escore de qualidade e FASTQ



ASCII
American Standard Code for Information Interchange
Código Padrão Americano para o Intercâmbio de Informação

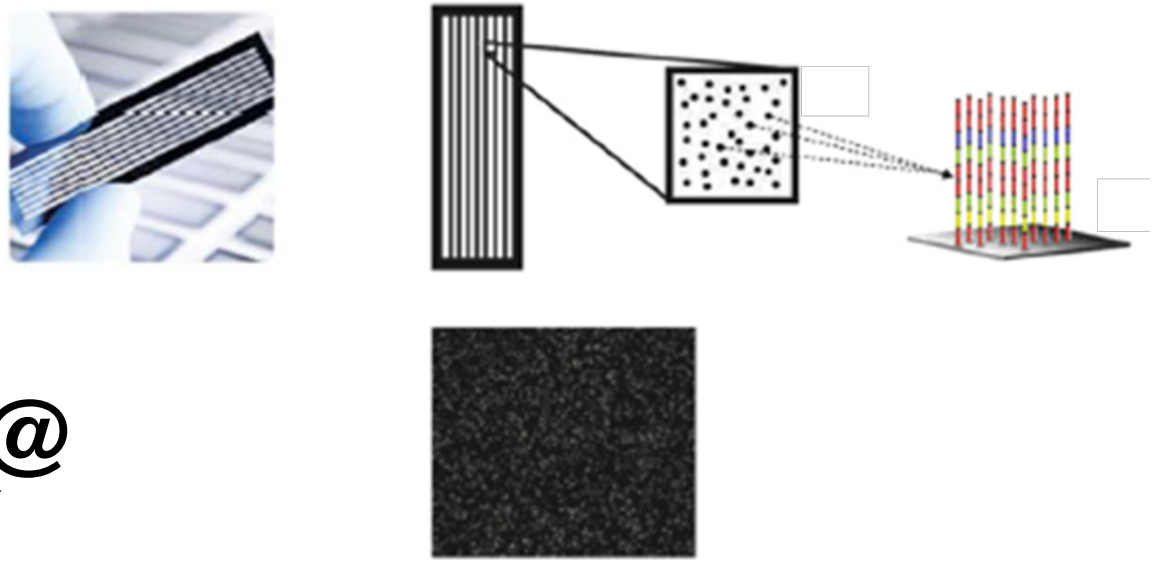
Codificação de caracteres de sete bits baseada no alfabeto inglês

Formato dos dados : Escore de qualidade e FASTQ

```
 SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....................................................
 ..........................XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX.....................
 ..............................IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII.....................
 ................................JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ.....................
 ..LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL.....................................................
 !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
 |                          |    |        |                              |                     |
33                        59   64       73                            104                   126
 0........................26...31.......40
                           -5....0........9.............................40
                                0........9.............................40
                                   3.....9.............................40
 0.2......................26...31........41

S - Sanger        Phred+33,  raw reads typically (0, 40)
X - Solexa        Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64,  raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64,  raw reads typically (3, 40)
    with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
    (Note: See discussion above).
L - Illumina 1.8+ Phred+33,  raw reads typically (0, 41)
```

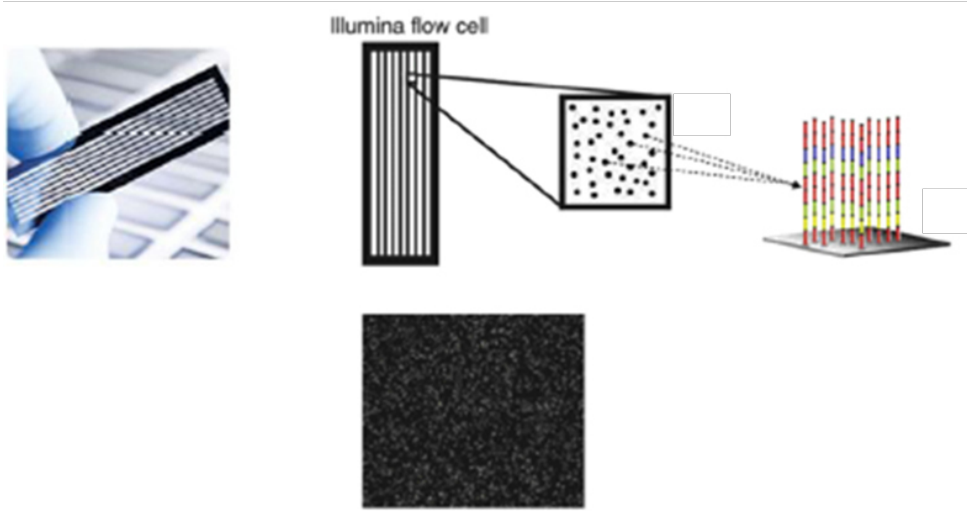
Formato dos dados : Exemplo de arquivo Illumina



@

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```

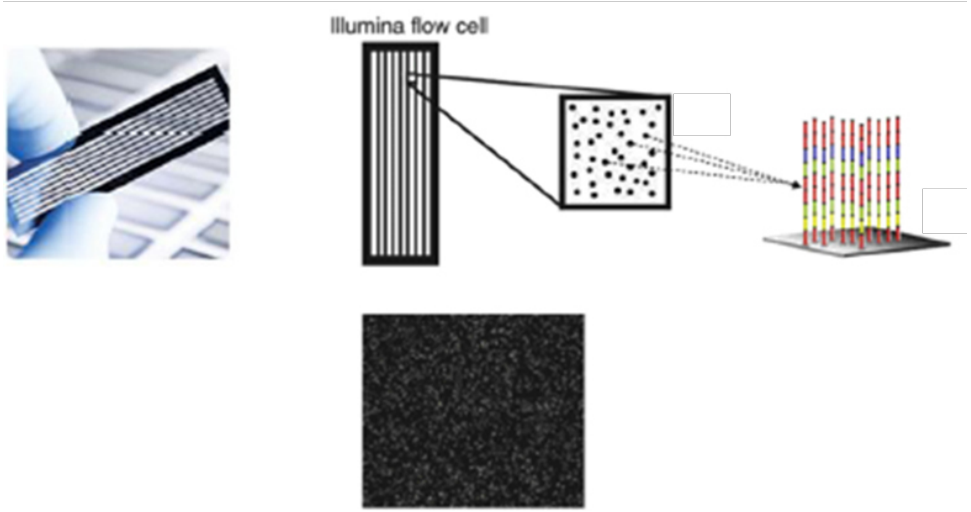
Formato dos dados : Exemplo de arquivo Illumina



Nome do sequenciador (= número de série)

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```

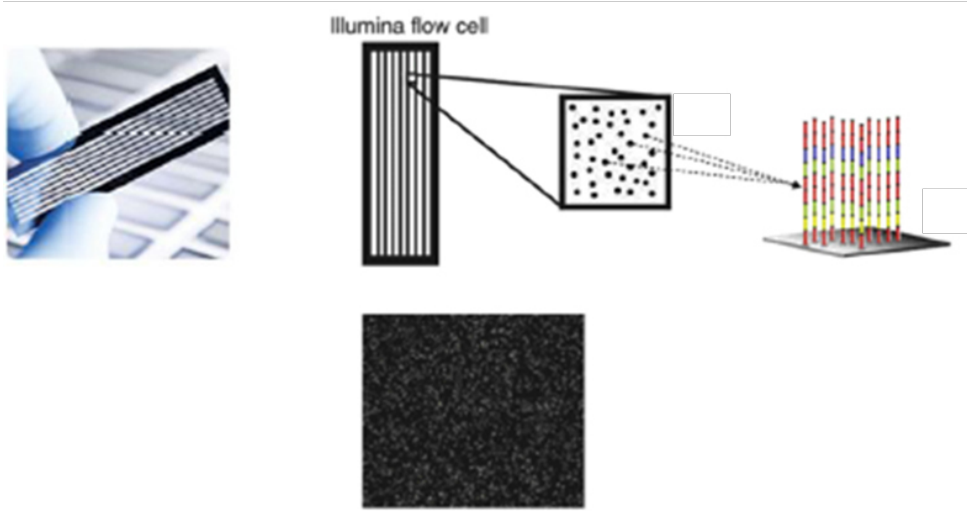
Formato dos dados : Exemplo de arquivo Illumina



Número da corrida

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```

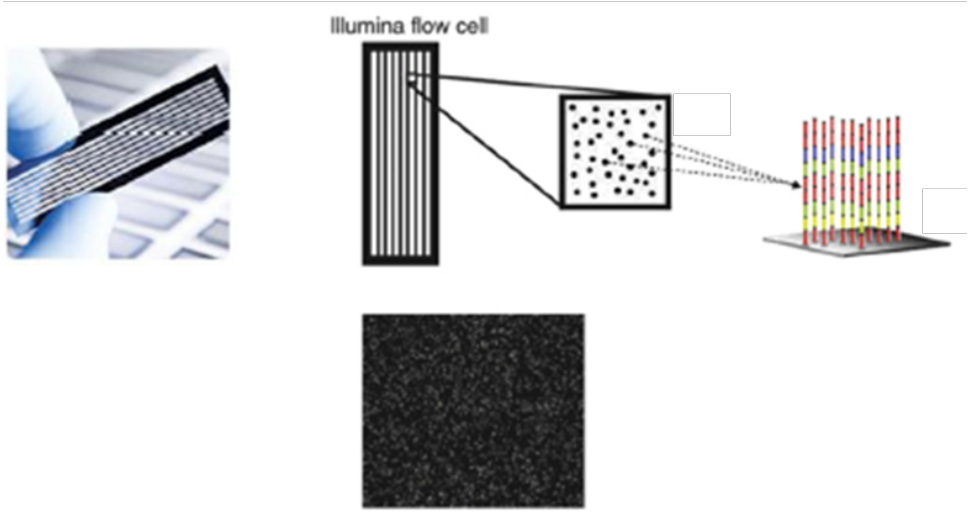
Formato dos dados : Exemplo de arquivo Illumina



ID da flowcell (lâmina)

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```

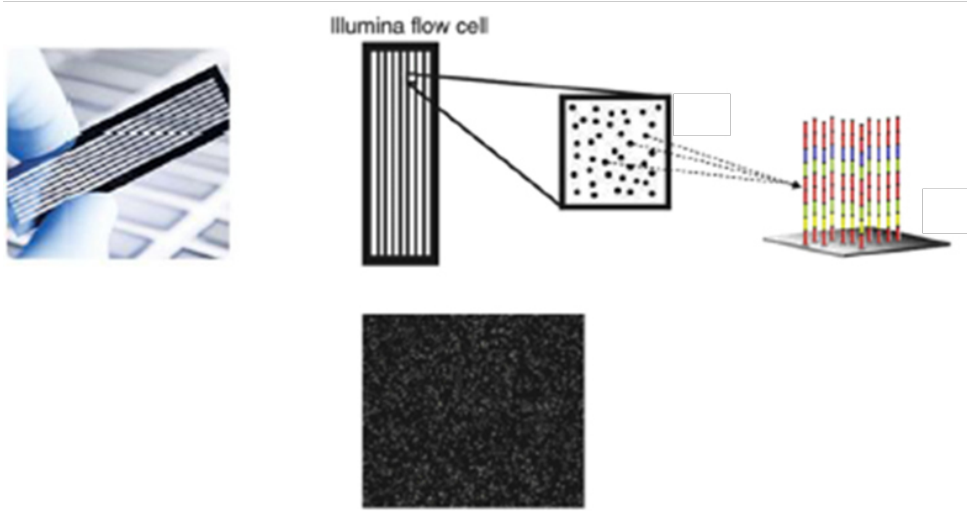
Formato dos dados : Exemplo de arquivo Illumina



Número da canaleta e do poço

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```

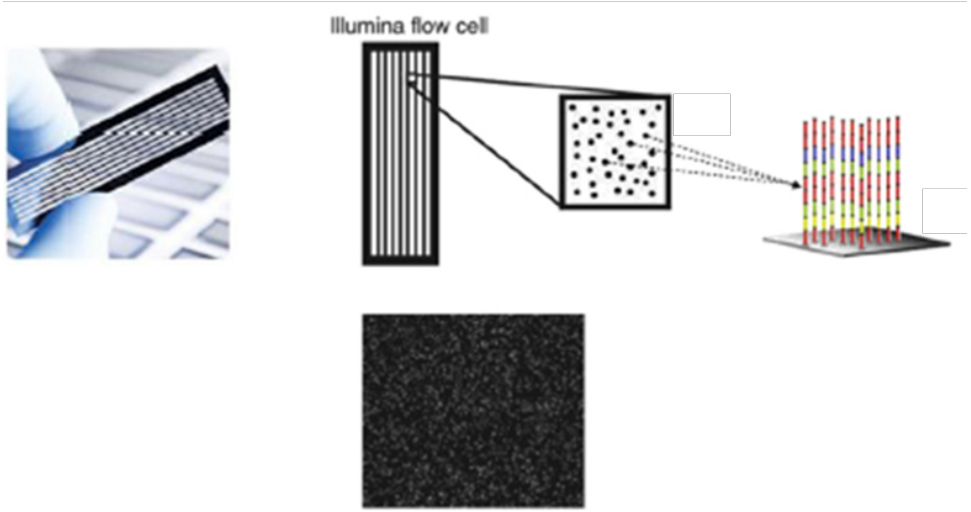
Formato dos dados : Exemplo de arquivo Illumina



Coordenada x e y do cluster no poço

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```

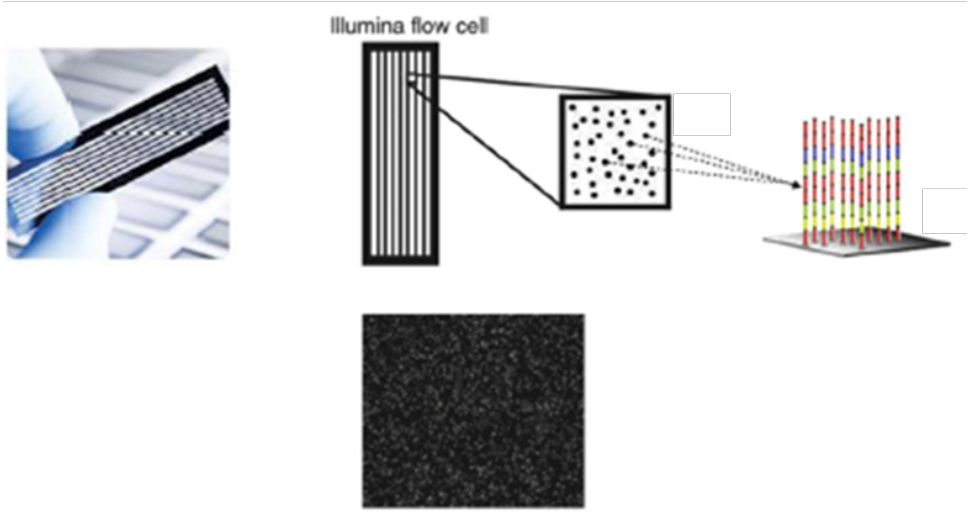
Formato dos dados : Exemplo de arquivo Illumina



Número do read (1 ou 2 se pair-end):read filtrado ou não:número de controle

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```

Formato dos dados : Exemplo de arquivo Illumina



Sequência do índice usado na amostra

```
@MG00HS20:1055:CARGWANXX:1:1101:4978:5857 1:N:0:GAGTGG
AGGTCTGTGCATCAAACATGGCAGAGGAAAGAGGTGTAAGGTGGAAGGCTGCACTCGTAGTGCGGAAGGA
CAGGTTGGGTTATGCATCTCTCATGGAGCAG
+
BBCBBFFGEGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGCGGGGFGGGGGGGGGGGGGGGGGFGGGGG
GGG=0E=ACFDGGGGFGGGGGGGGGG>FGGG
```